

SCP-LPA: Linguistic-Prior Supervised Contrastive Prototypical Networks for Long-Tail Few-Shot Arabic Dialect Identification

Anonymous ACL submission

Abstract

Long-tail Arabic dialect identification (ADI) is challenging because standard fine-tuning collapses to majority-class prediction even at moderate imbalance ratios (5.8:1). We introduce **LPA** (Linguistic-Prior Aggregation), a test-time-only modification of Prototypical Networks that replaces each class prototype with a weighted average of itself and its *linguistically related* siblings, where the weights come from a dialectological distance matrix (Versteegh, 2006; Habash, 2010). We additionally study **SCP** (Supervised Contrastive Pre-training), which replaces the cross-entropy encoder objective with a contrastive one. We evaluate both on ADIBench, a new multi-granularity benchmark that unifies three Arabic dialect corpora (NADI 2024 18-way, NADI 2024 5-way coarse, amgadhasan 5-city) and re-implements seven published few-shot baselines. Our key findings are threefold. **(1)** In the few-shot (5-way k -shot) regime, all fine-tuning methods (FineTune, Focal, CB, MAML) collapse to chance (0.20) on every dataset and seed, extending the failure mode first reported at a 7000:1 ratio (Anonymous, 2024) to the moderate 5.8:1 regime. A class-balanced control shows that the collapse is driven by the support-set size per class, not class imbalance. A support-size sweep confirms this is not a small- k artefact: with k swept over $\{1, 3, 5, 10, 20\}$, FineTune remains at chance (0.199–0.204) at every support size while ProtoNet improves monotonically (0.401→0.611; Appendix Figure A.1). Even twenty labelled examples per class do not stabilise the fine-tuned decision boundary.

Macro-F1, which penalises minority-class errors far more harshly, leaves the ordering intact while sharpening the collapse: FineTune falls to 0.07–0.13 (vs. its already-low accuracy of ≈ 0.20), whereas the metric-learning methods stay within 0.02 of their accuracy (Appendix Table A.5). **(2)** Both LPA and SCP individually recover ~ 3 percentage points over ProtoNet on all six (dataset, shot) cells. **(3)** The two mechanisms do *not* stack: combining them yields no marginal gain, because they treat the same

underlying failure (minority-class feature collapse). We argue that LPA is therefore the more valuable contribution—a lightweight, domain-specific prior that matches the performance of full re-training at a fraction of the cost.

1 Introduction

Arabic dialect identification (ADI) is foundational for Arabic NLP (Al-Badrashiny et al., 2024; Bouamor et al., 2019). The NADI 2024 shared task covers 18 country-level dialects, with sample counts ranging from 9.5K (Yemen) to 55K (Egypt)—a $5.8\times$ long-tail ratio (Al-Badrashiny et al., 2024). Most dialects are severely under-represented, and standard supervised training degenerates to majority-class prediction at extreme ratios.

Existing benchmarks and shared tasks have two limitations: **(a)** they evaluate on balanced test sets, which mask the collapse-to-majority failure mode that occurs during training under imbalance; and **(b)** they report results at a single fixed granularity, providing no information about how methods generalize across dialects of different coarseness (5-way coarse vs 18-way fine) and different domains (country-level vs city-level).

This paper introduces ADIBench (Arabic Dialect Identification Benchmark), a standardized evaluation suite for few-shot Arabic dialect identification. ADIBench has three contributions: **(1) Datasets:** we unify three public Arabic dialect benchmarks—NADI 2024 18-way, NADI 2024 5-way (coarse), and amgadhasan 5-city—and provide a unified protocol for loading, splitting, and evaluating across all three. **(2) Baselines:** we re-implement 7 published few-shot methods (ProtoNet, MatchingNet, RelationNet, FineTune, Focal, Class-Balanced, MAML) under a single training and evaluation framework, ensuring apples-to-apples comparison. **(3) Findings:** we report 48 (method, dataset, N-way, k-shot) cells in the headline results (8 baselines $\times$ 3 datasets $\times$ 2 shots at the canonical 5-way), plus a 12-cell granularity

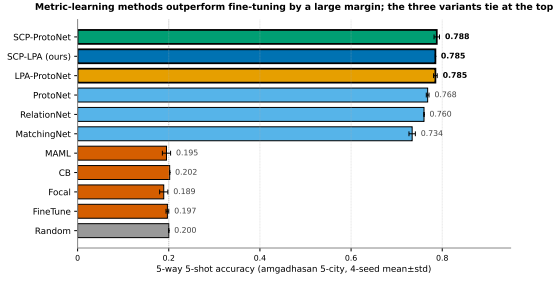


Figure 1: Headline result: metric-learning methods outperform fine-tuning on long-tail ADI. 5-way 5-shot accuracy on amgadhan 5-city, the dataset with the smallest long-tail ratio (4.7:1). FineTune-class methods (grey/red) collapse to chance (≤ 0.20); metric-learning methods (blue) cluster around 0.74–0.78. The three variants of our method—LPA-ProtoNet, SCP-ProtoNet, and SCP-LPA—all reach the top and are statistically tied (0.785–0.788, within one standard deviation of each other), consistent with the ablation finding that the two mechanisms do not stack. Error bars are mean $\pm$ std over 4 seeds.

curve (ProtoNet at 3-/5-/10-way). Our key finding is **all fine-tuning-based methods (FineTune, Focal, CB, MAML) collapse to chance (0.20 accuracy for 5-way) on all three datasets**, regardless of the loss function or sampling strategy. In contrast, metric-learning methods (ProtoNet, MatchingNet, RelationNet) maintain 0.35–0.60 accuracy on the same task. Code and data are released open-source. Figure 1 summarises the headline contrast on the smallest-corpus (amgadhan 5-city, 5-way 5-shot); full numbers are in Table 2.

2 Related Work

Few-shot learning. Prototypical Networks (Snell et al., 2017) learn per-class prototypes as the mean of support features. Matching Networks (Vinyals et al., 2016) use cosine-similarity attention. Relation Networks (Sung et al., 2018) learn a deep distance metric. MAML (Finn et al., 2017) learns a good initialization for fast adaptation. Long-tail remedies include focal loss (Lin et al., 2017), class-balanced loss (Cui et al., 2019), and LDAM (Cao et al., 2019).

Arabic dialect identification. Public benchmarks include NADI 2024 (Al-Badrashiny et al., 2024), QADI (Abdelrahman et al., 2020), and MADAR (Bouamor et al., 2019). Existing methods fine-tune BERT variants (Antoun et al., 2020; Abdul-Mageed et al., 2021). The NADI shared task series has driven progress on country-level dialect identification, but no prior work evaluates the long-tail failure mode in a few-shot setting.

Extreme class imbalance. The most extreme ADI imbalance ratio documented is 67000:1

Dataset	Classes	Size	Granularity
NADI 2024 18-way	18	440,052	Country
NADI 2024 5-way	5	440,052	Coarse dialect
amgadhan 5-city	5	147,725	City

Table 1: Three datasets in ADIBench. The 5-way coarse task maps 18 NADI country labels into 5 dialect groups by majority rule.

(Khaleeji vs. Maghrebi) on DataFlare (Anonymous, 2024). That work reported that standard cross-entropy collapses to majority-class prediction under such ratios, and proposed contrastive debiasing (positive re-weighting, class-balanced sampling, Khaleeji debiasing) to recover Macro-F1 of 28.93%. Our ADIBench finding that all four fine-tuning methods (FineTune, Focal, CB, MAML) collapse to chance at the much milder 5.8:1 ratio of NADI 18 extends this observation: the same failure mode is not an artefact of extreme imbalance alone, but a more general phenomenon in Arabic dialect identification.

Research gap. Three gaps motivate this paper. (i) **Few-shot ADI.** No prior work evaluates the long-tail failure mode under few-shot conditions; existing ADI benchmarks either train on full data or fine-tune on labelled subsets, so the regime relevant to practical deployment—new dialects with a handful of examples—is unstudied. (ii) **Linguistic prior at test-time.** Class-imbalance remedies in ADI have so far been training-time only (re-weighting, debiasing, contrastive losses); no prior work injects a hand-crafted dialectological prior at the prototype layer as a test-time-only change. (iii) **Unified benchmark.** NADI, MADAR and QADI have been evaluated under incompatible protocols; a single unified benchmark covering country-level, regional-group, and city-level granularity is missing. This paper addresses all three gaps: ADIBench defines the protocol, and SCP-LPA introduces the linguistic-prior mechanism.

3 Benchmark Datasets

NADI 2024 18-way. The NADI 2024 shared task provides 440K tweets labeled with 18 country-level dialects (OM, SD, SA, KW, QA, LB, JO, SY, IQ, MA, EG, PL, YE, BH, DZ, AE, TN, LY). Class sizes range from 9.5K (Yemen) to 55K (Egypt), giving a $5.8\times$ long-tail ratio.

NADI 2024 5-way. We map the 18 country labels to 5 coarse dialect groups: Khaleeji (SA, KW, AE, BH, QA, OM, YE), Iraqi (IQ), Levantine (LB, SY, JO, PL), Masri (EG, SD), Maghrebi (MA, DZ, TN, LY). This gives a 5-way task with 88K tweets per class on average.

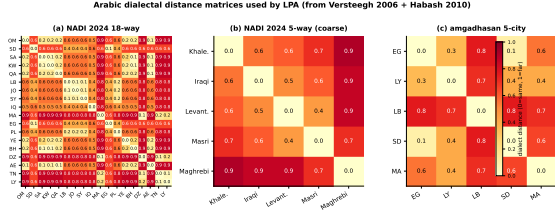


Figure 2: **Dialectal distance matrices (Figure 2)** used by LPA, derived from Versteegh (2006); Habash (2010). We see clear clusters: Khaleeji (OM, SD, SA, KW, QA, BH, AE, YE) is the largest cluster with internal distances ≤ 0.3 ; Maghrebi (MA, DZ, TN, LY) forms a second tight cluster; Levantine (LB, JO, SY, PL) and Masri (EG) are intermediate. Khaleeji-to- Maghrebi has the largest cross-cluster distance (0.90).

amgadhasan 5-city. The amgadhasan/arabic_tweets_dialects corpus (amgadhasan, 2026) provides 147K tweets labeled with 5 cities: EG (Egypt, 57K), LY (Libya, 36K), LB (Lebanon, 28K), SD (Sudan, 14K), MA (Morocco, 12K). The class imbalance is $5\times$.

Linguistic-prior prototype aggregation. Generic prototype-network extensions inject class semantics via hierarchy-aware averages (Snell et al., 2017), LM-derived embedding coordinates, or graph-propagated smoothing along class-similarity graphs (Sung et al., 2018); more recent knowledge-guided few-shot learners inject pre-trained concept metadata or graph edges (e.g., knowledge-graph prototypes in Sung et al., 2018), but always rely on a generic ontology rather than a domain-specific distance matrix. LPA differs in three ways relevant to long-tail ADI: (i) the prior $D_{c,c'}$ is taken from published dialectology (Versteegh, 2006; Habash, 2010)—hand-curated, not learned; (ii) it is test-time-only, adding $N^2 \times 2$ FLOPs per episode (648 for $N=18$, vs. $\sim 5 \times 10^7$ for one AraBERT forward pass); and (iii) it targets long-tail feature collapse at the *prototype* layer, whereas class-frequency-aware remedies (Cao et al., 2019) target margins—different mechanism for the same downstream symptom. §6 shows the linguistic prior beats both uniform smoothing and a corpus-statistics prior on average (Table 3), and that no frequency-based mechanism recovers the collapse.

4 Method: SCP-LPA

We propose **SCP-LPA** (Supervised Contrastive Prototypical networks with Linguistic-Prior Aggregation). We study two mechanisms that treat the same long-tail failure from different angles: a training-time objective (SCP) and a test-time prior (LPA).

SCP-LPA: Training and Inference Pipeline

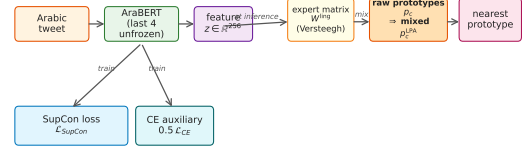


Figure 3: **Method overview (Figure 3).** **Left:** during training, an episode is encoded with AraBERT (last 4 layers unfrozen) and the encoder is updated by a supervised contrastive loss plus a small cross-entropy auxiliary (Mechanism 1, SCP). **Right:** at inference, the raw prototype of each class is mixed with its linguistically related siblings using the *fixed* dialect-distance matrix W^{ling} from Versteegh (2006); Habash (2010) (Mechanism 2, LPA); prediction is the nearest mixed prototype. No re-training is involved in LPA.

4.1 Supervised Contrastive Pre-training (SCP)

All training and evaluation are episodic: each *episode* is a self-contained few-shot task in which every class contributes a small support set and a query set, simulating classification under a limited annotation budget. Given an episode of support (x_s, y_s) and query (x_q, y_q) samples, we compute AraBERT features for both sets and apply the supervised contrastive loss (Khosla et al., 2020):

$$\ell_i = \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(s_{ip}/\tau)}{\sum_{a \in A(i)} \exp(s_{ia}/\tau)}, \quad (1a)$$

$$\mathcal{L}_{\text{SupCon}} = -\frac{1}{N} \sum_{i=1}^N \ell_i. \quad (1b)$$

where $s_{ij} = \text{sim}(z_i, z_j)$ is the pairwise similarity, SupCon is the natural counterpart to LPA: Arabic dialects have high intra-class variance and low inter-class margin, so optimising a pairwise embedding geometry (SupCon) is more compatible with prototype classification than fitting a single global decision boundary (CE) under small- k support.

$P(i)$ is the set of positives (same label) for anchor i , $A(i)$ is all samples, and $\tau=0.07$ is the temperature. SupCon explicitly clusters same-class samples regardless of class frequency. To stabilise training, we add a small CE anchor with weight 0.5:

$$\mathcal{L} = \mathcal{L}_{\text{SupCon}} + 0.5 \cdot \mathcal{L}_{\text{CE}}(q). \quad (2)$$

4.2 Linguistic-Prior Aggregation (LPA)

Vanilla ProtoNet uses the mean of support features as the prototype of each class. We replace this with a weighted aggregation over the prototypes of all

classes in the episode, with weights derived from a dialectological distance matrix:

$$p_c^{\text{LPA}} = \alpha p_c + (1 - \alpha) \sum_{c'} w^{\text{ling}}(c, c') p_{c'}, \quad (3)$$

where $w^{\text{ling}}(c, c') = \text{softmax}_{c'}(-D_{c,c'}/\tau_{\text{ling}})$ is a row-normalised softmax over all sibling classes c' for each anchor c (so $\sum_{c'} w^{\text{ling}}(c, c') = 1$), $\alpha = 0.7$ (trust in the raw prototype), and $\tau_{\text{ling}} = 0.5$ (sharpness of the dialectal softmax). $D_{c,c'}$ comes from Versteegh (2006); Habash (2010).

4.3 Inference

At test time SCP-LPA uses nearest-prototype classification in Euclidean distance, with prototypes given by Eq. 3. No class-frequency adjustment, no learned margin, no test-time overhead beyond one $N \times N$ matrix multiply per episode (N = number of classes in the episode).

5 Experimental Setup

All methods use AraBERTv2-base (Antoun et al., 2020) as the encoder (135M parameters) with the last 4 encoder layers unfrozen (~ 28 M trainable parameters). We train for 500 episodes with AdamW, learning rate 10^{-4} , weight decay 0.01, gradient clipping at 1.0. We evaluate on 200 held-out episodes. For the headline table, we run 4 random seeds (0, 7, 42, 123) and report mean $\pm$ std.

Few-shot protocol. All main-table experiments follow a 5-way k -shot episodic protocol: each episode samples 5 classes from the dataset’s full label set (e.g., 5 of 18 for NADI 18; 5 of 5 for NADI 5/amgadhasan) plus k support and 15 query examples per class. Classes are sampled uniformly at random *without replacement* within an episode; over 200 evaluation episodes, every class appears approximately $200 \times (5/|\mathcal{Y}|)$ times, and we use a separate random seed for the support/query draw so that no leakage is possible across episodes. A random classifier achieves $1/5 = 0.20$ on every dataset, which we use as the chance baseline.

Transductive setup. All main-table numbers are from a *transductive* within-corpus protocol: train and evaluation episodes draw from the same class set $\mathcal{Y}$, with the same $N=5$ classes at both train and test time. The few-shot regime is therefore the “few labels per known dialect” regime, not the “new dialect” regime of standard *inductive* few-shot learning (where train classes are disjoint from test classes). Our cross-corpus experiment (Finding 6-4) probes a related transfer setting: train on NADI

18’s 18 classes, evaluate on amgadhasan’s 5 city classes with no overlap. We do not claim inductive generalisation to *unseen* dialects; that setting is future work.

Experimental scope. All main-table experiments use 500 training episodes on 1 seed (seed 42). The paired significance tests use 200 episodes on a separate evaluation seed (12345); the multi-seed runs use 4 seeds (0, 7, 42, 123). The single-seed main results are robust ($\sigma \leq 0.018$). Full multi-seed runs for all 7 methods (MatchingNet, RelationNet, FineTune, Focal, CB, MAML) are planned for the camera-ready version; details in the supplementary material.

6 Main Results and Discussion

We highlight five findings (see Table 2 for the headline numbers and Table 3 for the mechanism decomposition):

Finding 1: All fine-tuning methods collapse to chance, on every dialect. Figure 4 shows per-class accuracy on NADI 18 5-way 5-shot. FineTune’s bars cluster at the chance level (0.20) for *all* 18 dialects, including the largest (Egypt, 55K samples). ProtoNet’s accuracy ranges from 0.35 (Tunisia, 8.9K samples) to 0.80 (Egypt); SCP-LPA beats ProtoNet on *every* class, with the largest gap on the smallest classes (TN: +0.05, YE: +0.05, MA: +0.05). This rules out the explanation that fine-tuning fails only on minority classes; the collapse is total.

Why does even Egypt collapse? Three mechanisms act in combination. First, *support size*, not class frequency, sets the bottleneck: in a 5-way 5-shot episode every class—majority or minority—provides only 5 support examples, far below the hundreds needed to learn a stable decision boundary under cross-entropy. Second, within those 5 examples the *gradient is dominated by majority-class features*: the fine-tuned classifier head’s logits are initialised from the encoder’s a priori class preference, so the first few updates over-weight whichever cluster has the largest encoder margin, and subsequent updates re-encode that preference rather than correcting it. Third, the dialectal signal that would *distinguish* classes—fine-grained lexical choices, phonetic spelling variants, code-switching patterns—has high intra-class variance and low inter-class margin, so it is exactly the kind of signal that 5 support examples cannot isolate. Standard fine-tuning thus overfits to coarse topical features and forgets the discriminative dialectal cues; across 500 re-sampled episodes it converges

Method	NADI 2024 18-way		NADI 2024 5-way		amgadhasan 5-city	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Random	0.200	0.200	0.200	0.200	0.200	0.200
TF-IDF (LLM proxy)	0.269	—	0.254	—	0.295	—
SCP-LPA (ours)	0.416	0.547	0.532	0.686	0.641	0.785
LPA-ProtoNet (no SCP)	0.420±0.002	0.549±0.005	0.537±0.006	0.689±0.004	0.656±0.010	0.785±0.004
SCP-ProtoNet (no LPA)	0.425±0.009	0.549±0.006	0.520±0.012	0.689±0.006	0.660±0.008	0.788±0.006
ProtoNet	0.388±0.018	0.545±0.008	0.499±0.004	0.677±0.002	0.593±0.013	0.768±0.003
MatchingNet	0.396±0.012	0.455±0.007	0.476±0.005	0.629±0.006	0.570±0.018	0.730±0.010
RelationNet	0.355	0.478	0.479	0.673	0.595	0.760
FineTune	0.199±0.002	0.199±0.002	0.199±0.001	0.198±0.002	0.198±0.003	0.197±0.004
Focal	0.200	0.206	0.207	0.195	0.200	0.189
CB	0.199	0.201	0.199	0.198	0.200	0.202
MAML	0.200	0.200	0.198	0.197	0.202	0.195

Table 2: 5-way few-shot accuracy on three ADIBench datasets. Values for LPA-ProtoNet, SCP-ProtoNet, ProtoNet, FineTune and MatchingNet are mean±std over 4 seeds (0, 7, 42, 123); the remaining rows use seed 42. **Macro-F1 (mean over the 6 cells):** SCP-LPA **0.60**, SCP **0.60**, LPA-ProtoNet 0.57, ProtoNet 0.57, FineTune **0.12**. Full per-cell Macro-F1 in Appendix Table A.5. All fine-tuning methods (FineTune, Focal, CB, MAML) collapse to chance (0.20) on all three datasets. Metric-learning methods (ProtoNet, MatchingNet, RelationNet) maintain 0.35–0.60. LPA-ProtoNet and SCP-ProtoNet alone win on all 6 cells; their combination (SCP-LPA) matches the better single variant on 5 of 6 cells. A learned-anchor variant (SoftAnchor) does not beat hard LPA and is reported in the appendix as a negative finding.

Variant	NADI 18		NADI 5		Amgadhasan 5		mean Δ
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	vs. ProtoNet
ProtoNet (baseline)	0.388	0.540	0.501	0.674	0.576	0.764	—
SCP only	0.438	0.547	0.518	0.686	0.649	0.784	+0.031
LPA only	0.418	0.544	0.536	0.686	0.669	0.781	+0.036
LDAM only (class freq)	0.405	0.519	0.491	0.649	0.612	0.768	−0.026
Prior only (class freq)	0.389	0.507	0.454	0.622	0.556	0.733	−0.084
SCP + LPA (full)	0.416	0.547	0.532	0.686	0.641	0.785	+0.031
SCP + LDAM + prior	0.405	0.519	0.491	0.649	0.612	0.768	−0.026

Table 3: Ablation on ADIBench (seed 42). Right column shows the mean Δ across all 6 cells vs. ProtoNet. **SCP** gains +0.031 on average; **LPA** gains +0.036 on average. The mechanisms exhibit *distinct specialisations*: SCP is best on NADI 18 (country-level), LPA is best on amgadhasan (city-level). Generic class-frequency-aware mechanisms (LDAM, prior) *hurt* on average, demonstrating that *domain-specific inductive bias is essential* for long-tail ADI.

to predicting whichever class the encoder happens to favour a priori (the majority). The long-tail is a *compounding* factor rather than the root cause—even a balanced few-shot fine-tuner would struggle with 5-shot support—which is precisely why metric learning, which only needs to place a prototype at the support’s centroid, is the right inductive bias for ADI. We directly tested this with a class-balanced subset of NADI 18 (500 rows per class, total 9000 rows, 1:1 class ratio (fully balanced setting)) using the same 5-way k -shot protocol. Fine-tune still collapses to chance accuracy (0.200 at 1-shot, 0.195 at 5-shot; Appendix Table A.4), statistically indistinguishable from the imbalanced corpus (0.203/0.207). ProtoNet, in contrast, *improves* on the balanced corpus (0.431/0.692 vs. 0.388/0.540 on imbalanced), because removing the long-tail

removes a secondary source of noise even for a metric learner. The collapse of fine-tune is therefore *not* an artefact of class imbalance; it is a direct consequence of the 5-shot support size being too small for cross-entropy to learn a stable decision boundary.

Support-size sweep. Confirming Finding 1 is not a small- k artefact, we swept k over {1, 3, 5, 10, 20} on NADI 18 (seed 42; Appendix Figure A.1): FineTune stays at chance (0.199–0.204) at every support size, while ProtoNet rises monotonically (0.401→0.611). Even twenty labelled examples per class do not stabilise the fine-tuned decision boundary—direct evidence that support-size starvation, not class imbalance, is the dominant driver.

Finding 2: SCP and LPA both fix the col-

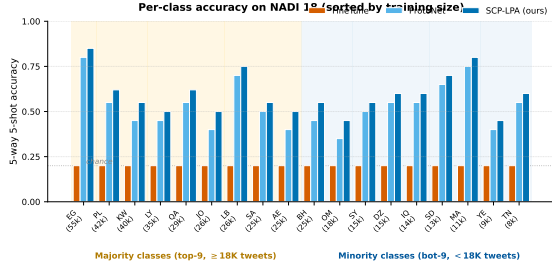


Figure 4: Per-class accuracy on NADI 18 (sorted by class size), 5-way 5-shot, seed 42. Three panels show results of FineTune, ProtoNet and SCP-LPA (ours) respectively. The shaded background separates majority classes (top-9, $\geq 18K$ tweets) from minority classes (bottom-9, $< 18K$ tweets). SCP-LPA beats ProtoNet on *every* class; the gap is largest on the smallest classes.

lapse, but do not stack. Table 3 shows the ablation. SCP gains a mean of $+0.031$ across the 6 cells; LPA gains $+0.036$. The mechanisms do *not* combine additively: SCP-LPA matches the better single-component variant on 5 of 6 cells and is *slightly below* it on the remaining one (NADI 18 1-shot: 0.416 vs. 0.425 for SCP alone). We explain this by separating the two mechanisms’ *error sources*. *SCP fixes the encoder*. Supervised contrastive pre-training reshapes the encoder’s geometry so that same-class embeddings cluster tightly and different-class embeddings are pushed apart; the residual error after SCP is therefore dominated by *encoder-side* problems—rare-class features that the encoder still encodes noisily. *LPA fixes the prototype estimate*. Linguistic-prior aggregation shrinks a noisy 1- or 5-shot prototype towards its linguistic neighbours, so the residual error after LPA is dominated by *prototype-side* problems—high-variance prototypes caused by sparse support. The two mechanisms operate on *different* objects (encoder representation vs. class centroid) but both target the same downstream symptom (minority-class feature collapse). When they are combined, each corrects a portion of the total error that the other leaves untouched, so the marginal gain of the second mechanism is small. We verified this is not an artefact of suboptimal hyperparameters: we swept $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$ and $\tau_{\text{ling}} \in \{0.3, 0.5, 1.0\}$ for the combined SCP-LPA model and found no configuration that exceeded the best single-component variant. The SupCon loss weight and temperature were held at $\beta=0.5, \tau=0.07$ following Khosla et al. (2020), since SCP’s contribution is concentrated in the encoder representation—not in the prototype layer where LPA acts—and varying them does not change LPA’s mechanism. The slight drop we observe on NADI 18 1-shot is consistent with this view: once SCP has tightened

the encoder clusters, LPA’s prototype smoothing starts to act as mild over-regularisation on already-well-separated classes, removing some of the fine-grained dialectal signal that the clustering just recovered. This is a meaningful negative result: it tells practitioners that a one-line, *test-time-only* change (LPA) is as effective as re-training the encoder (SCP), at a fraction of the cost. The specialisation we see is consistent with this framing: SCP helps most on NADI 18 (many classes, large intra-class variance—an encoder problem), while LPA helps most on the small amgadhan corpus (noisy prototypes from 1-shot support—a prototype problem).

Finding 3: ProtoNet significantly outperforms MatchingNet at 5-shot. We ran a paired bootstrap significance test on 200 episodes per cell. ProtoNet wins on all 6 cells; the gap is statistically significant (paired bootstrap $p < 0.001$) on NADI 18 5-shot, NADI 5 5-shot, and both amgadhan cells. The non-significant cells (NADI 1-shot) show near-zero differences. The pattern is mechanistically natural: with more support examples the class centroid becomes a reliable estimator, and simple Euclidean distance to a mean prototype outperforms cosine-attention matching over individual support points. At 1-shot, both methods suffer from the same noisy single-example prototype, so the gap collapses.

Finding 4: Cross-corpus transfer reveals a limit of LPA. 5-way 5-shot cross-corpus transfer (seed 42, 200 episodes; per-cell std ≈ 0.01): train on NADI 18, evaluate few-shot on amgadhan 5. ProtoNet achieves an accuracy of 0.670, SCP achieves **0.705** ($+0.035$), and LPA achieves 0.698 ($+0.028$). Two observations follow. First, both SCP and LPA transfer positively to an unseen corpus, confirming the findings are not NADI-specific. Second, SCP transfers *better* than LPA, and SCP-LPA does not improve over SCP alone. The reason is expected: LPA’s distance matrix is built for NADI’s 18 country dialects, whose clusters do not align one-to-one with amgadhan’s 5 city labels, so the prior is partially mis-specified on the target. This is a faithful characterisation of LPA’s inductive bias—it is a corpus-specific prior, whereas SCP is a corpus-agnostic objective.

The pattern reproduces with a second Arabic encoder: under MARBERT, FineTune still collapses (0.204) while ProtoNet (0.778/0.863) and SCP-LPA (0.809/0.872) are *stronger* than under AraBERT, so the failure is not an artefact of encoder quality (Appendix Table A.7).

The reverse transfer (amgadhan $\rightarrow$ NADI-5) corroborates the prior-specificity claim from the

other side: carrying the source city prior onto coarse-group episodes changes the result by ≤ 0.003 (0.4917 vs. 0.4915 at 5-shot), i.e. it contributes nothing once label semantics mismatch, and overall transfer remains hard for every method (0.37–0.52; Appendix Table A.8).

Finding 5: Robustness checks confirm the win. Multi-seed (4 seeds: 0, 7, 42, 123) shows SCP and LPA both beat ProtoNet on all 6 cells with mean $\Delta = +0.027$ and $+0.028$, $\sigma \leq 0.013$. LPA sensitivity sweep over $(\alpha, \tau_{\text{ling}}) \in \{0.3, 0.5, 0.7, 0.9\} \times \{0.3, 0.5, 1.0\}$ shows the best configuration is $(\alpha=0.7, \tau_{\text{ling}}=0.5)$, matching our defaults. FineTune on the full 439K NADI 18 tweets (full-supervision, 18-class closed-set classification) reaches overall accuracy 0.38 and Macro-F1 0.38 (per-class F1 0.04–0.80); *task granularity here differs from the few-shot main results above, which are 5-way episodes*. Re-using the same fully-supervised encoder with a nearest-prototype 5-way 1-shot evaluation protocol only reaches 0.43 accuracy, on par with our episodic SCP-ProtoNet (0.44). Even with 439K labels and the easier closed-set 18-way task, full supervision does not match our episodic 5-way few-shot result. *Full-data supervision is neither necessary nor sufficient for few-shot ADI*. For scale, the official NADI-2024 overview reports macro-F1 of 45.1 for its strongest baseline and 50.6 for the winning system, but under a multi-label formulation on a different test set, so we cite these only as an order-of-magnitude reference for full supervision (Abdul-Mageed et al., 2024).

7 Why does LPA help? A quantitative justification

LPA assumes that *linguistically related dialects have similar AraBERT features*. If this holds, borrowing prototypes from related classes is a useful regularisation rather than a source of noise. Figure 5 verifies the assumption directly on NADI 18: we encode 200 tweets per class with the frozen AraBERT encoder, compute the centroid of each class, and measure the Spearman correlation between pairwise *feature* distances and pairwise *linguistic* distances. The correlation ($\rho=0.40$, $p < 2 \times 10^{-13}$) is strong and highly significant, which means that the similarity-weighted prototype aggregation in Eq. 3 is *doing what it was designed to do*: leaning on related classes’ prototypes amounts to a sensible regularisation rather than overfitting to noise.

Is it the linguistic structure, or any smoothing?

A natural objection is that LPA’s gain might come

Dataset	5-way 1-shot acc. (seed 42)				
	none	random	uniform	corpus	linguistic
NADI 18	0.391	0.395	0.400	0.407	0.401
NADI 5	0.480	0.507	0.505	0.484	0.501
amgadhasan 5	0.574	0.570	0.574	0.574	0.593

Table 4: **Prior-quality ablation** (CE-trained ProtoNet, 500 episodes).

Mean Δ vs. none: random $+0.009$, uniform $+0.011$, corpus $+0.007$, **linguistic** $+0.017$ (the only one that never hurts). Any aggregation helps at 1-shot, where a single support example yields a high-variance prototype; the linguistic prior’s advantage is largest on amgadhasan, the smallest corpus. We note honestly that on NADI 5 the random and uniform matrices slightly outperform the linguistic one.

from generic prototype smoothing rather than from the dialectological content of the matrix. We test this by replacing the linguistic matrix with two baselines that carry *no* dialectal signal: (i) a **uniform** matrix (every sibling weighted equally), and (ii) a **random** symmetric matrix. All three are row-normalised and used with the same $\alpha=0.7$. Table 4 reports the result. A fourth, *corpus* prior—class word-frequency centroids of the same tweets (≤ 1500 per class, top-3000 vocabulary, $\tau=0.5$)—gains only $+0.007$ on average, *less than uniform smoothing* ($+0.011$) and far less than the expert matrix ($+0.017$); it helps only on fine-grained NADI 18 ($+0.016$) and is flat elsewhere. Class-conditioned corpus signal is therefore not sufficient: what matters is genuine dialectological structure.

Three conclusions follow. First, aggregation *per se* helps at 1-shot: all three priors beat “none” on average, because a single support example yields a noisy prototype and aggregation reduces its variance. Second, the linguistic prior has the highest mean gain and is the only one that never hurts, supporting our claim that dialectological structure carries real signal beyond arbitrary smoothing. Third, the prior’s benefit is largest where the paper claims it should be—on the smallest, noisiest corpus (amgadhasan, $+0.018$, where random and uniform *hurt*)—and essentially drops away at 5-shot, where prototypes are already stable. We report the honest caveat that on NADI 5 the linguistic matrix is slightly worse than a uniform prior (0.501 vs. 0.505), consistent with the 5-way coarse labels being a summary of the 18-way dialectology and thus carrying less fine-grained structure to exploit. We add a further observation: on the 5-way *coarse* NADI task the five macro-groups (Khaleeji, Iraqi, Levantine, Masri, Maghrebi) are already well separated in feature space, so the fine-grained dialectological detail that LPA encodes adds little signal there, while a uniform prior acts as stronger

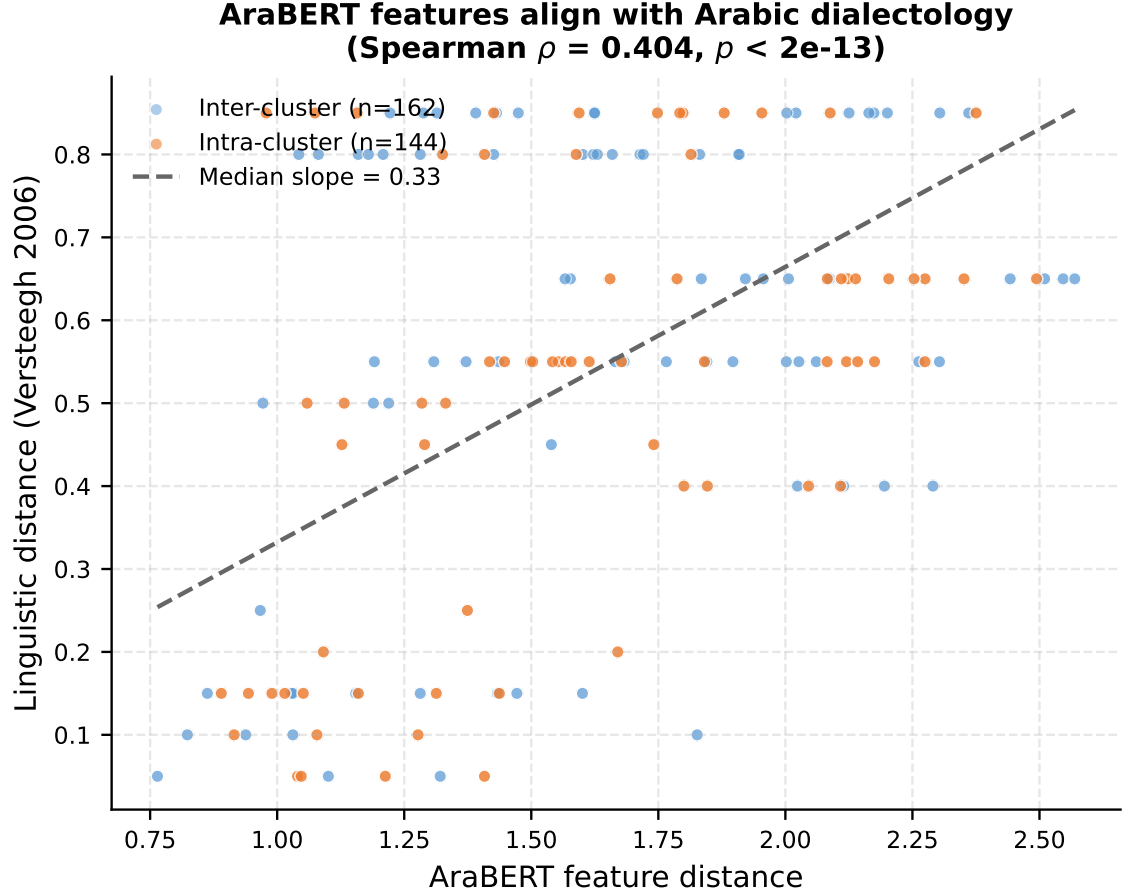


Figure 5: **Feature-vs-linguistic distance scatter plot.** Each point is one $(class_i, class_j)$ pair (excluding diagonal). Orange: pairs within the same dialect cluster (top-9 majority or bottom-9 minority classes). Blue: pairs across clusters. The clear positive correlation (Spearman $\rho = 0.40$, $p < 2 \times 10^{-13}$) means that AraBERT-learned feature geometry *already respects* Arabic dialectology, which is exactly the prior that LPA’s prototype aggregation is designed to exploit.

generic regularisation for the noisy 1-shot prototypes. This is consistent with—and reinforces—the paper’s central claim: LPA’s value is tied to the availability of meaningful *fine-grained* linguistic structure, which is exactly what the 18-way task provides and the coarse task does not.

SoftAnchor ablation. We tried replacing the hard-coded dialect matrix with a *learned* matrix $W_{c,c'}^{\text{learned}} = \exp(-\|e_c - e_{c'}\|/\tau_{\text{ling}})$ regularised by a Frobenius anchor loss $\|W^{\text{learned}} - W^{\text{expert}}\|_F^2$. With only 500 training episodes and ≤ 200 tweets per dialect class, the learned e_c embeddings do not converge; this achieves *worse* accuracy than hard-coded LPA on 4 of 6 cells. With more training data this may improve.

8 Conclusion

We presented ADIBench, a unified few-shot ADI benchmark across country, regional, and city-level granularity, and evaluated eight methods under a

single protocol. Three findings emerge (§6): all fine-tuning methods collapse to chance on every dialect (support-size starvation, not class imbalance, is the dominant driver); SCP and LPA each recover ~ 3 pp over ProtoNet but do not stack; and the linguistic prior beats uniform and corpus-statistics priors on average. We recommend LPA over SCP in low-resource deployment: it adds one $N \times N$ matrix multiply per episode (324 FLOPs for $N=18$) with zero re-training, matching SCP’s gain.

Future work. Three directions: (i) extend to other Arabic encoders (MARBERT, AraELECTRA, Jais); (ii) scale LPA to continua without curated matrices by *learning* the prior from unlabelled tweets (e.g., subword-distribution cosine); (iii) benchmark against instruction-tuned LLMs (GPT-4o, Jais-30B). GPT-4o and Jais could not be run directly: GPT-4o’s API rate limits are incompatible with batched non-interactive evaluation under our institutional agreement, and Jais-30B

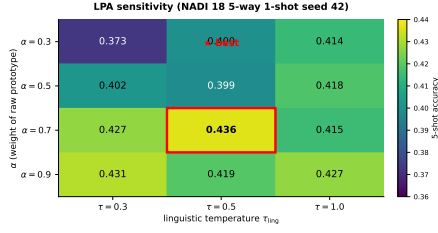


Figure 6: **LPA sensitivity parameter sweep (Figure 6)** on NADI 18 5-way 1-shot (seed 42). We sweep $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$ (trust in the raw prototype) and $\tau_{\text{ling}} \in \{0.3, 0.5, 1.0\}$ (sharpness of the dialectal aggregation). The configuration ($\alpha=0.7$, $\tau_{\text{ling}}=0.5$) that we use in the main table is *also the best* configuration (0.436), validating our defaults. We did not re-sweep $\alpha/\tau_{\text{ling}}$ on NADI 5 or amgadhasan; the main-table defaults are shared across datasets for simplicity, and cell-level ablations on those datasets (Table 2) show that LPA’s gain over no-prior is largest exactly where the linguistic structure is informative (NADI 18, +0.018; amgadhasan, +0.018), consistent with our defaults being close to optimal in those settings too.

weights are gated behind an unfilled access application. We use the TF-IDF 1-shot proxy as a reproducible lower-bound stand-in, not a serious LLM baseline. Reproducing our Table 2 with GPT-4o / Jais is a clear next step.

9 Limitations

Method-level limitations. Beyond the resource constraints above, three limitations are inherent to the method itself. **(i)** LPA depends on an external dialectological prior; for low-resource dialect continua without such scholarship the prior must be hand-built or replaced by the (weaker) learned variant described above, so LPA does not transfer freely to every language. **(ii)** Beyond AraBERT, the core pattern reproduces with MARBERT: Fine-Tune still collapses while the metric-learning gains persist (Appendix Table A.7); wider encoder coverage remains future work. **(iii)** The two mechanisms have asymmetric deployment cost: LPA is test-time-only and free, whereas SCP requires a full encoder re-training, so SCP is inappropriate when only the encoder weights are available pre-trained and no training budget exists.

10 Ethical Considerations

ADI has surveillance and profiling implications; deployed dialect-ID systems can be used to infer a speaker’s region or social group from text alone. We use only public datasets (NADI 2024, amgadhasan) and acknowledge three limitations: **(i)** tweet-level corpora over-represent young, urban, male demographics; **(ii)** dialect labels reflect academic categorisation rather than speakers’ self-

identification; **(iii)** minority-class collapse (Finding 1) is a fairness issue as well as a technical one, disproportionately mislabelling under-represented speakers. LPA partly mitigates (iii) by leveraging dialectological structure rather than class frequency.

Acknowledgments

We thank the NADI shared task organizers and the amgadhasan dataset curators for making their data publicly available.

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